Diagrama de clases de VeriFarmac

«layer»
Capa de Presentación (Flutter + Riverpod)

AuthNotifier «StateNotifier»
+signInWithGoogle(): void
+signOut(): void

ScannerNotifier «StateNotifier»
+scanBarcode(barcode: String): void
+scanOcr(text: String): void
+analyzeImage(imagePath: String): void
+reset(): void

MedicineNotifier «StateNotifier»
+loadByBarcode(barcode: String): void
+loadByRegistry(registry: String): void

HistoryNotifier «StateNotifier»
+load(): void
+delete(id: String): void
+deleteAll(): void
+filterBy(status: String): void
+sync(userId: String): void

DashboardNotifier «StateNotifier»
+load(): void

AuthState
+status: AuthStatus
+user: AppUser [0..1]
+error: String [0..1]

ScannerState
+status: ScannerStatus
+result: ScanResult [0..1]
+error: String [0..1]

MedicineState
+status: MedicineStatus
+medicine: Medicine [0..1]
+error: String [0..1]

HistoryState
+entries: List<HistoryEntry>
+isLoading: bool
+filter: String [0..1]

DashboardState
+stats: Map<String, int>
+isLoading: bool

1 (AuthNotifier → AuthState)
1 (ScannerNotifier → ScannerState)
1 (MedicineNotifier → MedicineState)
1 (HistoryNotifier → HistoryState)
1 (DashboardNotifier → DashboardState)

«use» (Capa de Presentación → Capa de Dominio)

«layer»
Capa de Dominio (Dart puro)

Casos de uso

Autenticación

SignOutUseCase
+call(): void

SignInWithGoogleUseCase
+call(): void

Escaneo

AnalyzeImageUseCase
+call(imagePath): ScanResult

ScanByOcrUseCase
+call(text): ScanResult

ScanByBarcodeUseCase
+call(barcode): ScanResult

Consulta INVIMA

SearchMedicineUseCase
+call(query): List<Medicine>

GetMedicineByBarcodeUseCase
+call(barcode): Medicine

GetMedicineByRegistryUseCase
+call(registry): Medicine

Historial

GetStatsUseCase
+call(): Map<String, int>

FetchHistoryUseCase
+call(page): List<HistoryEntry>

DeleteHistoryEntryUseCase
+call(id): void

SaveHistoryEntryUseCase
+call(entry): void

SyncHistoryUseCase
+call(userId): void

«use» (Casos de uso → Contratos de repositorio)

Contratos de repositorio

IAuthRepository
+usuarioActual: AppUser [0..1]
+signInWithGoogle(): void
+signOut(): void

IMedicineRepository
+findByBarcode(barcode: String): Medicine
+findByRegistry(registry: String): Medicine
+search(query: String): List<Medicine>

IScannerRepository
+processBarcode(barcode: String): ScanResult
+processOcr(text: String): ScanResult
+analyzeImage(imagePath: String): ScanResult

IHistoryRepository
+save(entry: HistoryEntry): void
+fetch(page: int): List<HistoryEntry>
+delete(id: String): void
+deleteAll(): void
+sync(userId: String): void
+getStatistics(): Map<String, int>

Entidades

AppUser
+id: String
+email: String
+nombre: String [0..1]
+avatarUrl: String [0..1]

ScanResult
+id: String
+scannedValue: String
+method: ScanMethod
+scannedAt: DateTime
+confidence: double
+medicineName: String [0..1]
+sanitaryRecord: String [0..1]
+status: String [0..1]
+error: String [0..1]

registra 0..1 — * corresponde a

Medicine
+id: String
+name: String
+sanitaryRecord: String
+laboratory: String
+condition: MedicineCondition
+holder: String [0..1]
+activeIngredient: String [0..1]
+concentration: String [0..1]
+pharmaceuticalForm: String [0..1]

HistoryEntry
+id: String
+medicineName: String
+sanitaryRecord: String
+status: String
+method: String
+createdAt: DateTime
+laboratory: String [0..1]
+confidence: double
+synced: bool

ScanMethod (E)
barcode
ocr
visual

MedicineCondition (E)
valid
expired
invalid
suspicious
unknown

«layer»
Capa de Datos

Repositorios

AuthRepositoryImpl
-dataSource: AuthRemoteDataSource

MedicineRepositoryImpl
-datasource: InvimaApiDataSource

HistoryRepositoryImpl
-local: HistoryLocalDataSource
-remote: HistoryRemoteDataSource

ScannerRepositoryImpl
-invima: InvimaApiDataSource
-ocr: OcrDataSource
-claude: ClaudeAiDataSource
-extractInvimaCode(text): String
-extractExpiryDate(text): DateTime
-isExpired(date): bool

Fuentes de datos

AuthRemoteDataSource «Supabase Auth»
+signInWithGoogle(): void
+signOut(): void

InvimaApiDataSource «Supabase»
+findByRegistry(registry): MedicineModel
+findByBarcode(barcode): MedicineModel
+search(query): List<MedicineModel>

HistoryLocalDataSource «SharedPreferences»
+save(entry): void
+fetch(page): List<HistoryEntryModel>
+fetchUnsynced(): List<HistoryEntryModel>
+markSynced(id): void

HistoryRemoteDataSource «Supabase»
+uploadEntries(entries): void
+downloadHistory(userId): List<HistoryEntryModel>
+delete(id): void

OcrDataSource «ML Kit»
+extractText(imagePath): String

ClaudeAiDataSource «Edge Function»
+analyzePackaging(imagePath): Map

«create»

Modelos (DTO)

AppUserModel
+fromSupabase(json): AppUserModel

MedicineModel
+fromJson(json): MedicineModel
+toJson(): Map

HistoryEntryModel
+fromJson(json): HistoryEntryModel
+toJson(): Map

ScanResultModel
+fromJson(json): ScanResultModel
+toJson(): Map

Presentación
Dominio
Datos